Proceedings of the International Conference on Sustainable Communication Networks and Application (ICSCNA 2023)
IEEE Xplore Part Number: CFP23DW8-ART; ISBN: 979-8-3503-1398-7

2023 International Conference on Sustainable Communication Networks and Application (ICSCNA) | 979-8-3503-1398-7/23/$31.00 ©2023 IEEE | DOI: 10.1109/ICSCNA58489.2023.10370715

Cognitive Stress Recognition during Mathematical Task and EEG changes following Audio-Visual Stimuli for Relaxation

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***Abstract* – Solving mathematical tasks or puzzles is one of the most common learning processes in many learning activities. This task can be either simple or complex depending on the level of study. Many individuals feel the fear of math tasks which will hamper the learning process due to stress. Hence, recognition of stress plays a crucial role in many learning activities. In this work, machine learning-based automatic recognition of mental stress during math puzzle solving is developed using EEG signals. EEG signals are recorded using Enobio 8 device in two categories, one during math puzzle solving and another in happy state with audio-visual stimuli with funny video clips. The raw signals are preprocessed to remove the artifacts using 2-40Hz band pass filter. Filtered data is used to extract EEG bands. EEG relative band power is analyzed to observe the changes during math task and relaxation. The two mental states are classified using Long Short Term Memory (LSTM) deep learning model. The classification accuracy of 86% is achieved in beta band.**

Index Terms* – Electroencephalogram, *Brain Computer Interface, Deep learning, Mental Arithmetic, Long Short Term Memory, Stress.

I. INTRODUCTION

In the fast-paced and demanding world of today, individuals often find themselves grappling with various forms of stress. One particularly intriguing facet of this phenomenon is cognitive stress, which emerges during complex tasks such as mathematical problem-solving. The human brain's response to stress during mathematical activities is a subject of growing interest, as it not only affects cognitive performance but also sheds light on the intricate interplay between the mind and mathematics [1]. Furthermore, the search for effective relaxation techniques has led to the exploration of innovative approaches, including audio-visual stimuli. These stimuli are believed to induce relaxation and have the potential to modulate brain activity as evidenced by quantitative EEG (Electroencephalogram) changes. This confluence of cognitive stress recognition during mathematical tasks and the influence of audio-visual stimuli on brain activity represents a fascinating intersection in the fields of psychology, neuroscience, and education [2].

Mathematics is a discipline that frequently elicits stress and anxiety in individuals of all ages. This phenomenon, often referred to as math anxiety, can have a significant impact on mathematical performance and learning outcomes. Understanding cognitive stress during mathematical tasks is vital for several reasons such as academic performance, psychological well-being and neurological implications. In recent years, there has been growing interest in leveraging technology to promote relaxation and stress reduction. One promising avenue is the use of audio-visual stimuli, which combine auditory and visual elements to induce relaxation. This research area is significant for stress management, neurofeedback etc. By understanding the neural correlates of stress during math-related activities and how audio-visual interventions can modulate brain activity, researchers can develop more efficient educational strategies, stress management tools, and therapies for individuals facing math-related stress and other forms of psychological distress. This research has the potential to improve learning outcomes, mental health, and overall quality of life for a broad range of individuals [1-2].

This exploration holds promise for understanding how people can use the power of technology to manage stress and optimize cognitive performance in educational and real-world environment. This research study will investigate the key aspects of cognitive stress recognition during mathematical tasks and the intriguing EEG changes that follow audio-visual stimuli for relaxation, by considering their potential implications for enhancing mental well-being and learning outcomes. Deep Learning based LSTM model is used to classify stress and relaxed states.

II. BACKGROUND

This section analyzes and summarizes some of the deep learning methods used in stress recognition and mental arithmetic task classification.

Lokesh M et al., [3] have proposed a hybrid method using Convolutional Neural Network (CNN) - Long Short-Term

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Memory (LSTM) for stress recognition. The classification accuracy of 99.2% is achieved using this hybrid model. Abhishek Varshney et al., [4] have proposed recurrent neural network based mental arithmetic task classification. The results showed good performance using LSTM, Bidirectional LSTM (BiLSTM), and Grated Recurrent Unit (GRU)-based RNN classifiers with accuracy of 99.81%, 99.43%, and 99.81% respectively.

Ganguly B et al., [5] have presented EEG classification using stacked LSTM for EEG based arithmetic task classification. M. A. Klados et al., [6] have presented automatic recognition of math anxiety using naïve bayes machine learning classifier. The classification accuracy of 93.75% is achieved using this classifier. U. Goenka et al., [7] have presented four deep learning methods for classification of EEG during math task. LSTM, BiLSTM, ConvLSTM and CNN-LSTM were used and ConvLSTM showed highest accuracy of 99.72% compared to other methods. B. Fatimah et al., [8] have proposed mental arithmetic task classification using support vector machine and frequency decomposition method. Kruskal-Wallis method is also used to select relevant statistical feature. K. Nirde et al., [9] have presented EEG mental arithmetic task levels classification using both machine learning and deep learning. K-Nearest Neighbour (KNN) in machine learning achieved accuracy of 91% and Artificial neural network in deep learning has shown accuracy of 96.80%. M. M. Rahman et al., [10] presented an end-to-end deep learning model for classification of mental arithmetic task. A residual-based temporal attention network (RTA-Net) is implemented for classification and it showed an accuracy of 99.32%.

Many papers in the current literature have demonstrated the classification of mental arithmetic tasks using deep learning methods. However, none of them have addressed relaxation techniques or presented EEG changes following relaxation methods. Therefore, this paper aims to investigate the cognitive stress of individuals during a mathematical task and how these levels can be modulated using audio-visual stimuli for relaxation.

III. METHODS AND MATERIALS

This section gives a detailed description of methodology for analysis and classification of EEG signals during math task and happy state.

A. Data Collection

The EEG signals were recorded using the Enobio-8 signal acquisition device. This device has 8-channels indicating 8-electrodes placed on the scalp. All participants were selected based on their normal mental health status, with ages ranging from 21 to 25 years. Subjects were presented with math puzzles, and an example is shown in Figure 1. The signals were recorded while participants solved the puzzles, and afterward, they were presented with a funny video to induce relaxation. EEG signals were also recorded after watching the funny videos. The mental state during the math task is considered as 'stress,' while the state after watching the funny video is considered as 'happy.' These two classes were used for analysis and classification.

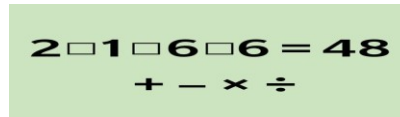

Fig. 1 Math Puzzle

B. Data Preprocessing

The raw EEG signals include artifacts that can originate from either physiological or non-physiological sources. Physiological artifacts occur internally and include phenomena like eye blinks, muscle movements, and deep breathing. Non-physiological artifacts, on the other hand, arise from external factors such as cable movements, electrode pops, and incorrect reference placement. These artifacts were removed using Parks-McClellan optimal finite impulse response (FIR) filter, which operates within a frequency range of 2-40Hz, using MATLAB [11-12]. The processed EEG signal was then used to extract distinct EEG frequency bands, including delta, theta, alpha, beta, and gamma, utilizing a multi-wavelet transformation technique. In this multi-wavelet transformation, we employed the Daubedhies 14 wavelet to decompose the filtered signal across six levels [11]. EEG relative band power is calculated for each EEG bands such as alpha, beta, delta, theta and gamma. Subsequently, we individually considered each EEG band to extract statistical features. These statistical features were standardized using the standard scaler feature scaling method, which ensures that the features have a mean of 0 and a standard deviation of 1.

C. Classification using Deep Learning method

The feature scaled data is used for training the deep learning model. The deep learning method used is described as follows.

i) Long Short Term Memory (LSTM)

Long Short-Term Memory (LSTM) is a type of recurrent neural network (RNN) architecture designed to handle sequential data and capture long-range dependencies. It is particularly well-suited for tasks like time series prediction, natural language processing, and speech recognition.

At the start of the sequence, the LSTM cell needs initial values for its hidden state (h_0) and cell state (C_0).

- h_0=0 (Initial hidden state)
- C_0=0 (Initial cell state)

LSTM processes one element of the input sequence at a time (x_t). For each time step, it performs several operations to update its cell state and hidden state. The forget gate (f_t) determines what information from the previous cell state (C_{t-1}) should be discarded and what information should be retained. It takes the previous hidden state (h_{t-1}) and the current input (x_t) as input.

$$f_t = \sigma (W_f . [h_{t-1}, x_t] + b_f) \quad (1)$$

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The input gate decides what new information should be stored in the cell state. It takes the previous hidden state (h_{t-1}) and the current input (x_t) as input.

$$i_t = \sigma (W_i . [h_{t-1}, x_t] + b_i) \quad (2)$$

Candidate Cell State (C_T) is the new candidate values that could be added to the cell state. It is computed using the previous hidden state (h_{t-1}) and the current input (x_t).

$$C_T = \tanh (W_c . [h_{t-1}, x_t] + b_c) \quad (3)$$

The updated cell (C_t) state is a combination of the previous cell state (C_{t-1}) and the candidate cell state (C_T) based on the forget gate (f_t) and input gate (i_t).

$$C_t = f_t . (C_{t-1}) + i_t . C_T \quad (4)$$

After updating the cell state, the LSTM calculates the output at the current time step. The output gate (O_t) determines what information from the cell state (C_t) should be included in the hidden state (ht). It takes the previous hidden state (h_{t-1}) and the current input (x_t).

$$O_t = \sigma (W_o . [h_{t-1}, x_t] + b_o) \quad (5)$$

Hidden state (h_t) is computed by applying the output gate (O_t) to the cell state (C_t).

$$h_t = O_t . \tanh (C_t) \quad (6)$$

The hidden state (h_t) at each time step can be used for various tasks such as prediction, classification, or generating sequences.

Where,

x_t: Input at time step t.

h_t: Hidden state at time step t.

C_t: Cell state at time step t.

W_f, W_i, W_c, W_o: Weight matrices for the forget gate, input gate, candidate cell state, and output gate, respectively.

b_f, b_i, b_c, b_o: Bias terms for the corresponding gates.

σ: Sigmoid activation function.

tanh: Hyperbolic tangent activation function.

In this work, softmax is used as an activation function. Since it is classification problem, categorical_crossentropy loss function is used. The initial learning rate considered is 0.001 and number of epochs is 50. Early stopping is included in the implementation to stop the training when there is no significant change in the validation accuracy

IV. RESULTS

This section presents the results of EEG changes after audio-visual stimuli using funny clips and evaluation of LSTM method for classification of happy and stress mental states. The relative band power provides valuable information about the distribution of signal power across EEG frequency bands and is typically expressed as a ratio or percentage. Alpha band refers to relaxed state, beta refers to stress, delta refers to sleep, theta refers to drowsiness and gamma refers to peak performance. The relative band power is calculated for these bands for happy and stress states. Figure 2 shows the alpha relative band power error bars with standard error. Figure 2 clearly shows an increase in the band power in happy state. Increase of alpha band power indicates relaxation after watching funny clips. Figure 3 shows beta relative band power error bars with standard error. In beta band, stress percentage is higher. Figure 4 shows delta relative band power error bars with standard error. Figure 4 shows an increase in happy state. Figure 5 shows theta relative band power error bars with standard error. Figure 5 shows an increase in happy state. . Figure 6 shows gamma relative band power error bars with standard error. Figure 6 shows an increase in stress in gamma band. In this work, alpha band is considered vital as it offers relaxation. As per figure 2, there is a clear indication of relaxation due to increase in relative band power of happy state.

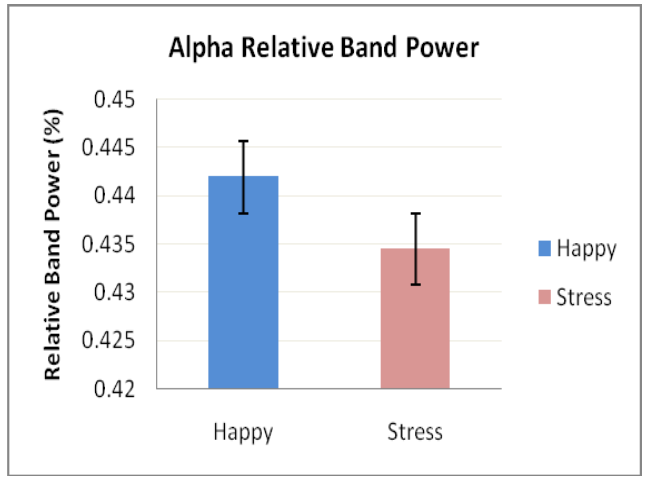

Fig. 2 Alpha Relative Band Power Error Bar

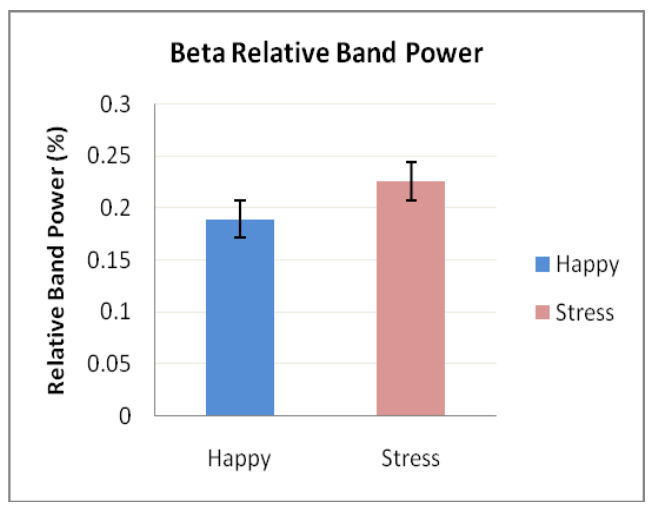

Fig. 3 Beta Relative Band Power Error Bar

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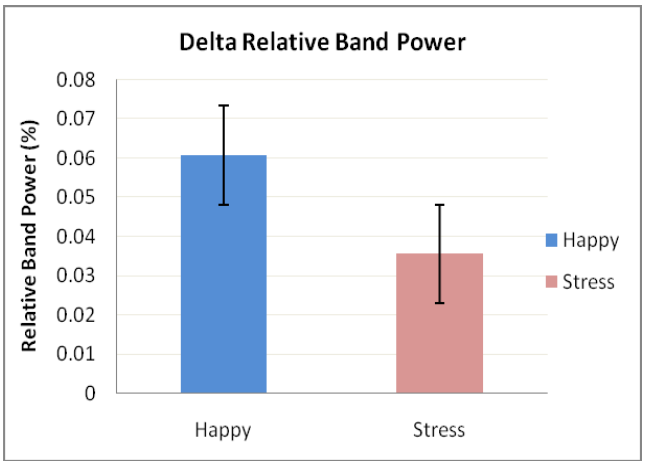

Fig. 4 Delta Relative Band Power Error Bar

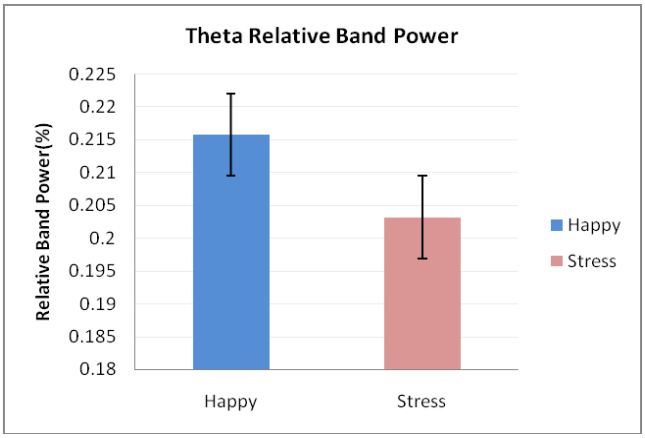

Fig. 5 Theta Relative Band Power Error Bar

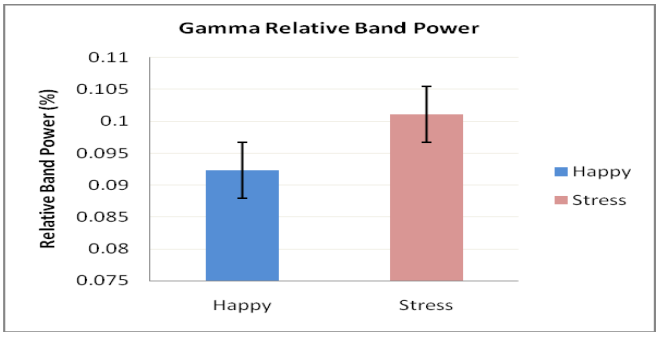

Fig. 6 Gamma Relative Band Power Error Bar

Table 1 presents the training and validation accuracy of LSTM model. The average training accuracy across all the bands is 71% and validation accuracy is 76%. In beta band validation accuracy is 86%. The accuracy can be improved further by using hyperparameter tuning.

Figure 7 shows the training & validation accuracy and training & validation loss in alpha band. Training loss is 0.6 and validation loss is 0.57. Figure 8 shows the training & validation accuracy and training & validation loss in beta band. Training loss is 0.55 and validation loss is 0.50. Figure 9 shows the training & validation accuracy and training & validation loss in delta band. Training loss is 0.51 and validation loss is 0.52. Figure 10 shows the training & validation accuracy and training & validation loss in theta band. Training loss is 0.57 and validation loss is 0.57. Figure 11 shows the training & validation accuracy and training & validation loss in gamma band. Training loss is 0.60 and validation loss is 0.55.

TABLE I
TRAINING AND VALIDATION ACCURACY FOR EACH EEG BAND

Sl No	EEG bands	Training Accuracy	Validation Accuracy
1	Delta	79%	71%
2	Theta	64%	71%
3	Alpha	72%	71%
4	Beta	74%	**86%**
5	Gamma	68%	79%
Average	**All**	**71%**	**76%**

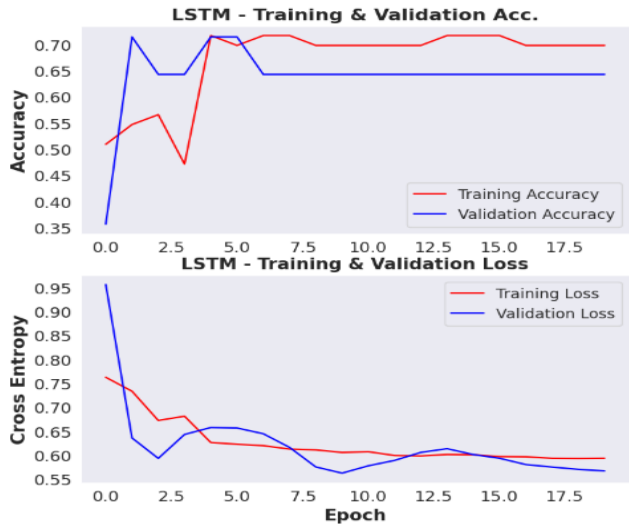

Fig. 7 Alpha Band LSTM Training & validation Accuracy and Loss

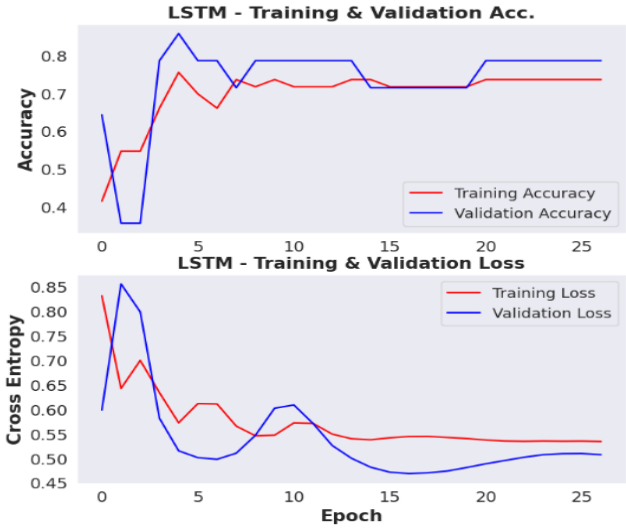

Fig. 8 Beta Band LSTM Training & validation Accuracy and Loss

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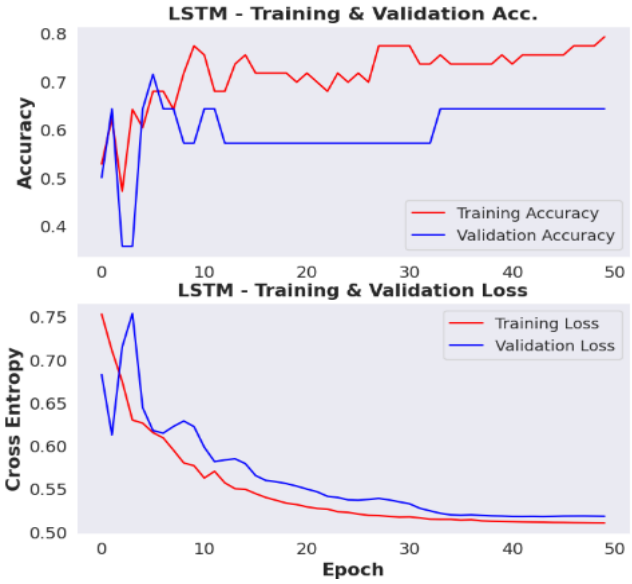

Fig. 9 Delta Band LSTM Training & validation Accuracy and Loss

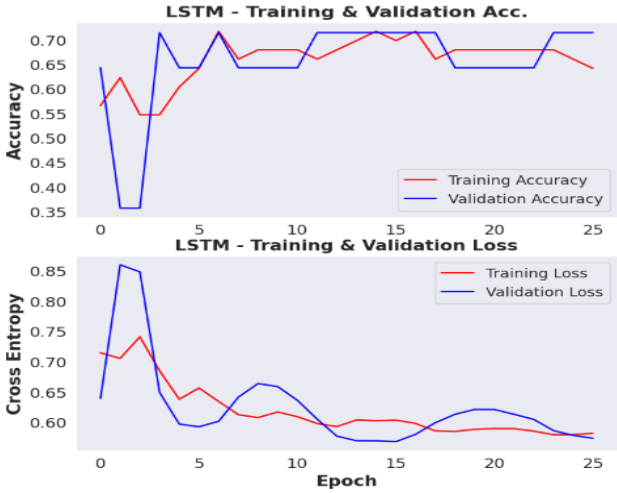

Fig. 10 Theta Band LSTM Training & validation Accuracy and Loss

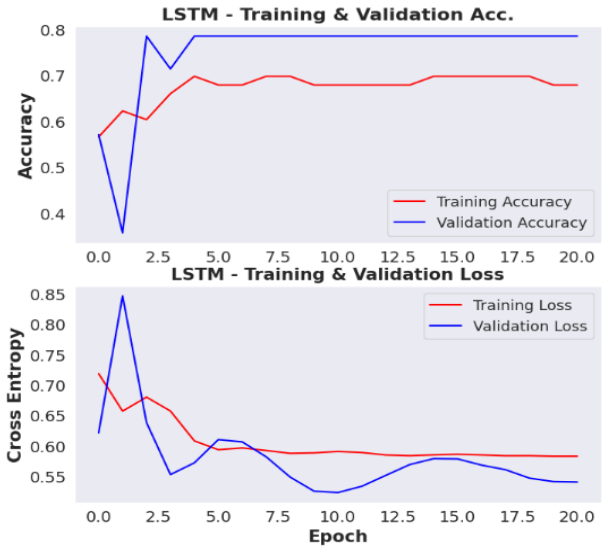

Fig. 11 Gamma Band LSTM Training & validation Accuracy and Loss

V. Conclusion

This work explores how our brains react when we're stressed while doing math and how we can use relaxing videos and technology to help. In today's busy world, understanding how stress affects us during math is crucial. This study reveals an increase in alpha relative band power during the 'happy' state, indicating relaxation. The classification of 'happy' and 'stressed' states using LSTM achieved an accuracy of 86% in the beta band. Furthermore, there is potential to enhance this system for improved accuracy and its application to other mathematical tasks.

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